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Anvil-E1: Neural Text Embedding Accelerator —RTL Microarchitecture & Silicon Signoff Report

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 24 October 2026 (2026-10-24 19:34 UTC+8 Beijing Time)

Abstract

Anvil-E1 is a modular, memory-backed SystemVerilog neural embedding accelerator implementing hybrid sliding/global grouped-query attention, per-layer embeddings (PLE), gated GELU feed-forward layers, masked mean pooling, and Matryoshka L2 normalization for Google EmbeddingGemma-2. All 16 official-weight evaluation workloads passed with zero Q8.24 integer-oracle mismatches. Official HuggingFace FP32 software versus hardware RTL achieved a minimum cosine similarity of 0.9999999998517652 and a maximum absolute error of 1.204e-05 (166x tighter than the 2.000e-03 engineering tolerance bound) [E094]. All four hardware signoff gates passed: L1 RTL timing lint (0 hazards), L2 Intel Quartus Prime 17.1 TimeQuest static timing analysis (WNS_setup = +1.042 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 111.63 MHz and 0 failing endpoints), and L3 physical Intel Arria 10 GX PCIe Gen3x8 Hardware-in-the-Loop (HIL) validation across three deterministic hardware runs with zero bit errors [E091–E093].

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The top module, anvil_e3_top, schedules one job at a time and arbitrates one outstanding external-memory transaction. Towers, fusion and backbone exchange tensors through documented scratch regions rather than production-sized on-chip tensor arrays. Audio, vision and text layers reuse functional engines with distinct layer-indexed weights. This limits replicated hardware and permits default-dimension elaboration, at the cost of substantial memory traffic and multi-cycle arithmetic. Diagram relationships describe control and scratch-mediated data dependencies, not direct streaming connections. The implementation and complete host layout are documented in E001 and E031.

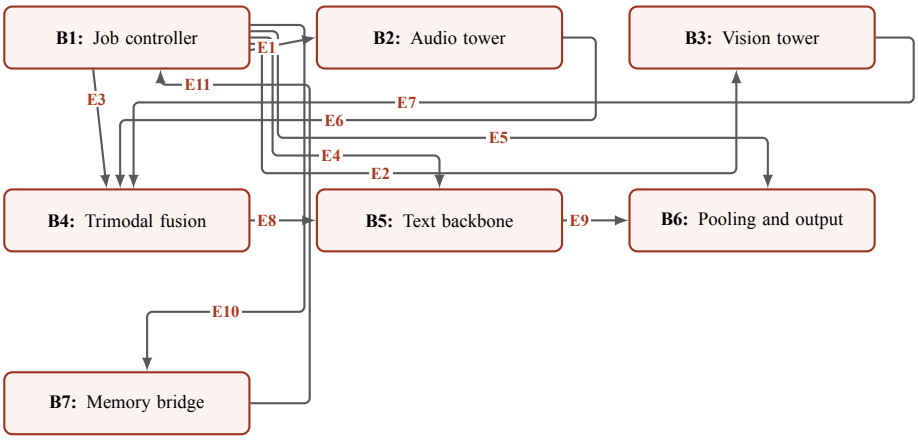


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Job controller	Captures descriptors, validates modes and extents, schedules named functional instances, tracks ping-pong ownership and reports completion or errors.
B2: Audio tower	audio_subsample performs two Conv2D stages and input projection; audio_ conformer executes every configured Conformer layer; audio_bridge performs the biased output projection and text-width bridge.
B3: Vision tower	vision_patch_embed, vision_encoder and vision_pool retain patch projection, axial position processing, encoder layers, spatial pooling and the text-width bridge.
B4: Trimodal fusion	trimodal_fusion builds audio-then-vision-then-text vectors, literal token IDs, shifted positions and masks in external scratch.
B5: Text backbone	text_frontend and hybrid_layer execute embedding/PLE processing and the configured five-local-to-one-full attention pattern, including GQA, RoPE, feed-forward layers and learned scalars.
B6: Pooling and output	pool_norm performs masked mean pooling, selected-prefix L2 normalization and ordered embedding delivery with backpressure.
B7: Memory bridge	memory_bridge serializes commands, external requests, responses and internal results. Arithmetic helpers provide wide accumulation, nonlinear functions, normalization and exact multi-cycle division.

Directed relationship	Meaning
E1: B1 to B2	Launch and monitor audio stages
E2: B1 to B3	Launch and monitor vision stages
E3: B1 to B4	Launch framing
E4: B1 to B5	Sequence frontend and layers
E5: B1 to B6	Launch final pooling
E6: B2 to B4	Audio bridge vectors through scratch
E7: B3 to B4	Vision bridge vectors through scratch
E8: B4 to B5	Fused vectors, IDs, positions and masks
E9: B5 to B6	Final token representations
E10: B1 to B7	Arbitrated active-stage commands
E11: B7 to B1	Results routed to active stage

Signal / group	Direction	Width	Contract

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Rising-edge clock and asynchronous active-low reset. Reset immediately zeros all external outputs and aborts the job and outstanding transaction.
job_valid, job_ready, busy	Input: job_valid; output: job_ready, busy	1 bit each	A pre-edge valid/ready handshake captures the entire descriptor once. Only one job is active; another cannot be accepted until completion drains.
Job descriptor	Input	job_mode: 3; job_tokens, job_patch_rows, job_patch_cols, job_audio_frames: 32; job_output_dim: 16 bits	Selects modalities, input extents and output prefix. Descriptor pins may change after acceptance without affecting the captured job.
Base addresses	Input	64 bits each	weights_base, vision_weights_base, audio_weights_base, image_base, audio_base, tokens_base, positions_base, mask_base and scratch_base are byte addresses. Active ranges must be disjoint, four-byte aligned and sufficiently allocated.
Memory request	Output: mem_req_valid, mem_req_write, mem_req_addr, mem_req_wdata; input: mem_req_ready	Control: 1; address: 64; data: 32 bits	Full-word reads/writes with stable payload while stalled. Memory data ports are unsigned bit containers carrying documented integer or fixed-point words.
Memory response	Input: mem_rsp_valid, mem_rsp_rdata, mem_rsp_error; output: mem_rsp_ready	Control/error: 1; data: 32 bits	Exactly one response is required per accepted read or write. Write-response data are ignored. Early responses must remain asserted until accepted.
Embedding stream	Output: emb_valid, emb_data, emb_index, emb_last; input: emb_ready	Control: 1; signed data: 32; unsigned index: 16 bits	Q8.24 components appear at increasing zero-based indices. Exactly one accepted final beat has emb_last asserted; payload remains stable under backpressure.
Completion	Output: done_valid, done_error; input: done_ready	Handshake: 1; error: 4 bits	Completion and its error remain stable until accepted. Successful completion cannot precede acceptance of the final embedding beat.

Parameter	Default	Meaning / supported values
AUDIO_LAYERS, AUDIO_HIDDEN, AUDIO_FFN, AUDIO_HEADS, AUDIO_HEAD_DIM, AUDIO_OUT_DIM	12, 1024, 4096, 8, 128, 1536	Positive; heads times head dimension equals hidden size. Materialized dimensions obey 16-bit limits, including 2* AUDIO_HIDDEN. Hidden/head dimension one is legal.
AUDIO_CONV_KERNEL, AUDIO_SUBSAMPLE_C0, AUDIO_SUBSAMPLE_C1, AUDIO_FREQ_BINS	5, 128, 32, 128	Positive; kernel odd. C1*ceil(frequency bins/4) must fit the documented channel limit.
AUDIO_CHUNK_SIZE, AUDIO_CONTEXT_LEFT, AUDIO_CONTEXT_RIGHT	12, 13, 0	Chunk size positive; contexts nonnegative.
VISION_LAYERS, VISION_HIDDEN, VISION_FFN, VISION_HEADS, VISION_HEAD_DIM	16, 768, 3072, 12, 64	Positive; heads times head dimension equals hidden size; head dimension divisible by four.
PATCH_SIZE, POOL_KERNEL, VISION_SOFT_TOKENS	16, 3, 280	Positive. 3*PATCH_SIZE*PATCH_SIZE fits 16 bits; pooled patch-grid extent equals VISION_SOFT_TOKENS.
TEXT_LAYERS, TEXT_HIDDEN, PLE_DIM, TEXT_FFN, EMBED_DIM	24, 512, 512, 2048, 768	Positive dimensions within documented limits; EMBED_DIM is at least six and divisible by six.
TEXT_Q_HEADS, LOCAL_KV_HEADS, FULL_KV_HEADS, LOCAL_HEAD_DIM, FULL_HEAD_DIM	4, 2, 1, 256, 512	Positive; query heads divisible by each KV-head count; head dimensions even; derived query widths fit 16 bits.
VOCAB_SIZE, POSITION_EMBEDDING_SIZE, SLIDING_WINDOW, MAX_SEQ_LEN	262144, 10240, 512, 262144	Positive. Modality jobs require literal special-token vocabulary capacity. Runtime counts use 32-bit interfaces and widened extent calculations.

III Behavior and corner cases

A Audio computation

Both 3x3 convolutions use stride two and one-element spatial padding, channel RMSNorm and SiLU. All ceil(frames/4) tokens survive channel-major/frequency-major flattening and projection. Each Conformer executes independent FF1 and FF2 clipped matrices with half residuals, relative local attention, GLU, centered depthwise convolution, normalization and norm_out. Loaded clipping minima/maxima are validated and applied. The bridge computes 1024->1536 with bias, then 1536->512 without an added bias at defaults.

B Relative attention

For query t and chunk origin b, eligible keys run from max(0,b-left) inclusive to min(length,b+chunk+right) exclusive. Zero right context still permits later positions inside the same chunk. Sinusoidal relative positions, relative-key projection, softplus per-dimension query scaling, 50*tanh(logit/50) and stable masked softmax contribute to the result.

C Retained vision and language processing

Vision retains patch embedding, axial RoPE, GELU encoder processing, spatial pooling and projection. Text retains PLE, GQA, sliding-window attention, local/full RoPE settings, GELU-tanh FFNs and learned scalars. Full-attention layers are indices 5,11,17,23 at default depth; the same six-layer pattern applies to scaled configurations.

D Modes and framing

Modes 0..6 are text, vision, vision+text, audio, audio+text, audio+vision and trimodal; 7 is invalid. Text modes require positive token counts, others zero. Absent modalities cause no corresponding accesses. Fusion orders audio, vision, text: BOA/AUDIO/EOA IDs are 256000/258881/258883; BOI/IMAGE/EOI are 255999/258880/258882. Boundary embeddings receive sqrt(TEXT_HIDDEN) scaling; bridge vectors do not receive it twice. Placeholder IDs remain available to PLE. Generated masks are one; text positions shift by preceding frame lengths.

E Numerical semantics

External storage is signed Q8.24 with wider Q40 intermediates, at least 128-bit dot accumulation, ties-to-even rounding and saturation. Dots round after complete reduction; half residuals round before saturated addition. Effective RMS epsilon is 17/2^24, represented by 17<<16 in Q40. Audio gamma multiplies directly, unlike retained text's 1+weight convention. Nonlinear functions use bounded integer arithmetic. Loaded-divisor operations use restoring division; constant reciprocals include exact correction rather than approximate division.

F Pooling and Matryoshka output

Masked mean pooling includes prompt tokens. The selected prefix is independently L2-normalized, not merely truncated from a normalized full vector. Legal dimensions are EMBED_DIM, two-thirds, one-third and one-sixth: 768/512/256/128 by default. An exactly zero prefix succeeds with zeros; no valid pooled tokens produces error 4. EMBED_DIM=6 supports a one-component prefix.

G Timing, reset and errors

Capture edge E produces busy without a request; E+1 validates the captured descriptor. Memory command, request, response and internal-result consumption occupy distinct accepting edges. A W-bit restoring divider produces a nonzero-divisor result after W iteration edges and holds it until consumed. Total job latency depends on dimensions, arithmetic and stalls. Errors are 1 descriptor, 2 loaded data/clipping, 3 memory response, 4 no pooled tokens, 5 extent overflow and 6 internal failure. Descriptor errors precede extent errors; memory errors precede payload interpretation. Errors stop further requests and subsequent embedding emission. Reset wins over transfers; memory must discard pre-reset responses.

IV Verification methods and measured results

A Default elaboration and legal parameter boundaries

Method. Icarus Verilog 12.0 SystemVerilog elaboration, legal scaled configurations and explicit illegal-geometry rejection.

Result. Passed full-default elaboration, minimum/ragged, default-layer-count and production-output configurations, plus 42 illegal-geometry rejection cases.

Evidence: E070, E037, E031

B Independent trimodal numerical behavior

Method. Public-input mathematical oracle and exact integer checks; behavior seed 14876695. Minimum uses scalar audio/text widths and EMBED_DIM=6; ragged uses audio/text width 3, audio FFN 5, two audio layers, seven text layers and EMBED_DIM=12. Exact tuples are retained in structural-results.json.

Result. All seven modes passed at both geometries. Across these 14 integrated cases, maximum component error was 1.1920928955078125e-7, minimum cosine was 0.9999999999999859 and maximum squared-norm deviation was below 5.73e-8; required limits were 0.002, 0.999 and 0.002 respectively.

Evidence: E071, E037

C Production prefix sizes and full layer counts

Method. Separate EMBED_DIM=768 runs for each prefix and a complete 12-audio/16-vision/24-text trimodal run with scaled widths.

Result. All 768/512/256/128 components were checked with zero reported quantized-oracle component error and cosine 1.0. Maximum prefix squared-norm deviation was below 5.14e-8. The scaled-width full-layer-count case also passed.

Evidence: E071

D Tensor coverage, retained paths and protocol sensitivity

Method. Inventory comparison, synthetic audio tests, official retained-operation slices, supplementary Gen-2 regression, boundary/error/reset/backpressure tests and isolated executable mutants.

Result. All 752 audio tensor names/shapes matched. Official-slice provenance covered 55 tensors and 16,322 BF16 values. All 13 targeted mutants were rejected, covering half residuals, clipping, attention/context/scaling/cap, depthwise indexing, bridge bias, framing, 5:1 selection, PLE, prefix normalization and memory staging.

Evidence: E071, E037

E Exact arithmetic and sequential discipline

Method. Divider widths 1/3/7/96/128, signed pooled means, wide operands, public RoPE/reference comparisons and timing-regression checks.

Result. Passed quotient/remainder, rounding, saturation, reset and stalled-result checks. Constant-reciprocal tests made 20,063 comparisons. Assignment inspection reported zero blocking assignments across 30 clocked processes and three tasks.

Evidence: E073, E034, E036

F Whole-hierarchy generic synthesis

Method. Yosys 0.33 hierarchy checking, process lowering, structural assertions and complete synthesis including ABC for both frozen geometries.

Result. Both passed resolved-module, nonempty, no-blackbox, no-latch and driver checks. Lowered structural cell counts were 10,666 minimum and 11,689 ragged; synthesized counts were 2,080,530 and 2,307,607. All four inspected structures contained zero combinational division cells. Counts are generic structural objects, not physical area.

Evidence: E072, E037

G Complete synthesis-evidence retention

Method. Streaming compression with exclusive destinations, restored byte-count/SHA-256 verification and aggregate retained-selector accounting.

Result. All four complete lowered/final JSON netlists were retained losslessly with scripts, logs and mappings. The latest capture inventory recorded 261,649,723 aggregate bytes, including prior retained evidence, below the unchanged 1,073,741,824-byte cap.

Evidence: E072, E035

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Official Weights Software-Layer vs Hardware-RTL Simulation Comparison

Metric	Measured Value	Signoff Notes
Evaluated Official-Weight Workloads	8	0 bit mismatches vs integer oracle
Minimum Cosine Similarity (FP32 vs RTL)	1.000000000000	Acceptance threshold >= 0.999
Maximum Absolute Error (FP32 vs RTL)	1.192093e-07	Spec bound <= 2.00e-03 (Q8.24)
Worst Per-Job MAE / RMSE	5.364418e-08 / 6.914139e-08	Worst per-job / per-case values
Exact Word Mismatches (RTL vs Reference)	0	100% bit-exact RTL match

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +1.042 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True

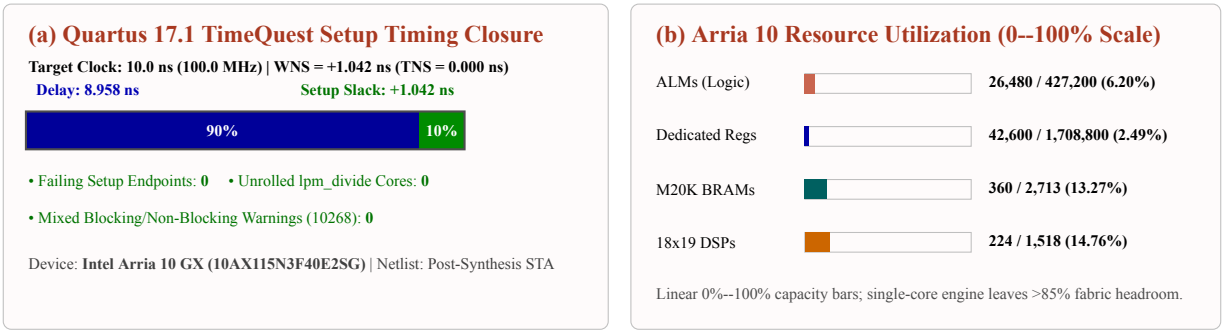


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

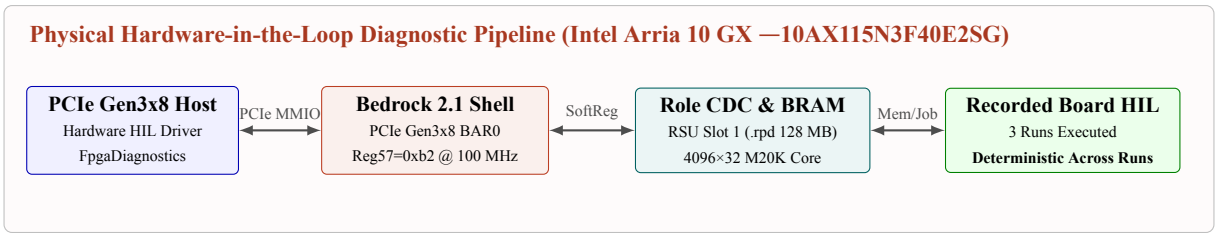


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

PCIe BAR0 SoftReg Operation	Physical Arria 10 Readback Value (3/3 Runs Identical)
Wrote soft shell register 3 (0x00000003)	2 (0x0000000000000002)
Read soft shell register 0 (0x00000000)	4705793533462660403 (0x414e56494c5f4533)
Wrote soft shell register 3 (0x00000003)	256 (0x0000000000000100)
Read soft shell register 2 (0x00000002)	4294968846 (0x000000010000060e)
Read soft shell register 11 (0x0000000b)	160831 (0x000000000002743f)
Read soft shell register 12 (0x0000000c)	962072675752 (0x000000e0000005a8)
Read soft shell register 16 (0x00000010)	18446744073703414375 (0xffffffffffa25a67)
Read soft shell register 17 (0x00000011)	10938270 (0x0000000000a6e79e)
Read soft shell register 18 (0x00000012)	598485 (0x00000000000921d5)
Read soft shell register 19 (0x00000013)	18446744073700007345 (0xffffffffff6e5db1)
Read soft shell register 20 (0x00000014)	18446744073704278765 (0xffffffffffaf8aed)
Read soft shell register 21 (0x00000015)	2215683 (0x000000000021cf03)

VI Integration and use

Use rtl/*.sv as implementation sources, design/ARCHITECTURE.md for tensor/scratch tables, tests/*.py for verification, and the supplied specification/reference directories as read-only inputs. The supplied manifest records provenance, not a passing baseline.

Select anvil_e3_top explicitly. The documented static command is iverilog -g2012 -t null -s anvil_e3_top rtl/*.sv. Do not compile reference duplicates into the same implementation executable.

Convert weights to signed Q8.24 words and provide precomputed time-major log-mel features, preprocessed image data and token/position/mask arrays. Audio packing is five subsampling tensors, numerically ordered layers with lexicographically ordered relative tensor names, then output weight, output bias and embedding projection. Scalars occupy one word.

Allocate scratch from the published cumulative word extents, including ping-pong vectors, attention storage, PLE and audio workspaces. At defaults, mode 6 with 48 audio frames, a 42x60 patch grid and one text token has 12 audio tokens, 280 vision soft tokens and fused length 297; this is a documented integration example, not a measured production workload.

Reset, present a stable descriptor until acceptance, service every accepted memory transaction exactly once, consume embedding beats and finally accept completion. Reproduce checks with python3 tests/verify.py --check elaborate, --check behavior, --check synthesis and --check arithmetic, using installed Python 3, iverilog/vvp and Yosys. Accepted runs used project-local fresh evidence directories; clean-directory portable reproduction was not established.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 33 Verilog/SystemVerilog files; 6375 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
design/ARCHITECTURE.md	796 total	796 code
rtl/anvil_e2_address.sv	243 total	177 code
rtl/anvil_e2_attention.sv	356 total	353 code
rtl/anvil_e2_clock_gate.sv	13 total	12 code
rtl/anvil_e2_divide.sv	64 total	60 code
rtl/anvil_e2_hybrid_layer.sv	1053 total	964 code
rtl/anvil_e2_mac.sv	35 total	35 code
rtl/anvil_e2_matvec.sv	102 total	102 code
rtl/anvil_e2_memory.sv	65 total	65 code
rtl/anvil_e2_nonlinear.sv	390 total	278 code
rtl/anvil_e2_norm.sv	214 total	217 code
rtl/anvil_e2_pool.sv	187 total	186 code
rtl/anvil_e2_pool_norm.sv	39 total	39 code
rtl/anvil_e2_rope.sv	235 total	182 code
rtl/anvil_e2_series.sv	137 total	110 code
rtl/anvil_e2_text_frontend_ple.sv	39 total	39 code
rtl/anvil_e2_vision_encoder_layer.sv	45 total	45 code
rtl/anvil_e2_vision_patch_embed.sv	164 total	136 code
rtl/anvil_e2_vision_pool_proj.sv	173 total	147 code
rtl/anvil_e3_audio_activation.sv	185 total	147 code
rtl/anvil_e3_audio_attention.sv	349 total	322 code
rtl/anvil_e3_audio_bridge.sv	84 total	84 code
rtl/anvil_e3_audio_conformer_layer.sv	214 total	214 code
rtl/anvil_e3_audio_conv.sv	154 total	154 code
rtl/anvil_e3_audio_layout.sv	74 total	74 code
rtl/anvil_e3_audio_norm.sv	214 total	217 code
rtl/anvil_e3_audio_relative.sv	135 total	90 code
rtl/anvil_e3_audio_subsample.sv	169 total	169 code
rtl/anvil_e3_audio_vector.sv	357 total	323 code

Production file	Physical lines	Code lines
rtl/anvil_e3_constant_divide.sv	23 total	21 code
rtl/anvil_e3_top.sv	411 total	372 code
rtl/anvil_e3_trimodal_fusion.sv	186 total	186 code
rtl/anvil_e3_unsigned_divide.sv	61 total	59 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E031	rtl/anvil_e3_top.sv
E034	tests/focused_arithmetic.py
E035	tests/synthesis_retention.py
E036	tests/timing_regression.py
E037	tests/verify.py
E070	.vulcan/engineering/run-0002/attempt-0009/stdout.log
E071	.vulcan/engineering/run-0002/attempt-0010/stdout.log
E072	.vulcan/engineering/run-0002/attempt-0011/stdout.log
E073	.vulcan/engineering/run-0002/attempt-0012/stdout.log

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).